

de Rham and Dolbeault theorems

Preliminaries

Definition. If $\mathcal{A}$ is a sheaf of rings, a sheaf $\mathcal{F}$ of $\mathcal{A}$ -modules is said to be a *sheaf of free $\mathcal{A}$ -modules* if there exists an integer n such that $\mathcal{F}$ is locally isomorphic to $\mathcal{A}^n$ as a sheaf of $\mathcal{A}$ -modules. The integer n is called the *rank* of $\mathcal{F}$.

Lemma. Let $\mathcal{A}$ be the sheaf of continuous, differentiable or holomorphic functions over a topological, differentiable or complex manifold X . For every vector topological, differentiable or holomorphic bundle $E \rightarrow X$, the presheaf $\mathcal{E}$

$$\mathcal{U} \mapsto \{\text{cont., dif. or hol. sections } \sigma : \mathcal{U} \rightarrow E|_{\mathcal{U}}\}$$

is a sheaf of modules over its corresponding sheaf of functions such that the correspondence

$$E \mapsto \mathcal{E}$$

is an equivalence of categories between vector bundles and sheaves of free $\mathcal{A}$ -modules.

Definition. • A *morphism of sheaves* $\phi : \mathcal{F} \rightarrow \mathcal{G}$ is a collection, for each open set $\mathcal{U}$, of morphisms $\phi_{\mathcal{U}} : \mathcal{F}(\mathcal{U}) \rightarrow \mathcal{G}(\mathcal{U})$ such that for $\sigma \in \mathcal{F}(\mathcal{U})$ and $V \subset \mathcal{U}$ we have $\phi_{\mathcal{U}}(\sigma)|_V = \phi_V(\sigma|_V)$.

- The morphism ϕ is *injective* (resp. *surjective*) if for every $x \in X$ the morphism

$$\phi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$$

is injective (resp. surjective).

- The presheaf $\mathcal{U} \mapsto \ker(\phi_{\mathcal{U}} : \mathcal{F}(\mathcal{U}) \rightarrow \mathcal{G}(\mathcal{U}))$ is a sheaf called the *kernel* of ϕ .
- The sheaf associated to the presheaf $\mathcal{U} \mapsto \text{img}(\phi_{\mathcal{U}} : \mathcal{F}(\mathcal{U}) \rightarrow \mathcal{G}(\mathcal{U}))$ is called *image* of ϕ .
- The sheaf associated to the presheaf $\mathcal{U} \mapsto \text{coker}(\phi_{\mathcal{U}} : \mathcal{F}(\mathcal{U}) \rightarrow \mathcal{G}(\mathcal{U}))$ is called *coker* of ϕ .

de Rham's theorem

Let's prove that singular cohomology groups are isomorphic to de Rham cohomology groups:

$$H_{\text{dR}}^q(X) \cong H_{\text{sing}}^q(X, \mathbb{R}).$$

References are from **voisin**. **Purple** means I'm unsure.

1.

Lemma (4.26). If an abelian category $\mathcal{C}$ has sufficiently many injective objects, then every object of $\mathcal{C}$ admits an injective resolution, i.e. a resolution $I^\bullet$ by a complex all of whose objects are injective objects.

2.

Theorem (4.28, Derived functors of a left-exact functor exist.). Let $\mathcal{C}$ and $\mathcal{C}'$ be two abelian categories, and let F be a left-exact functor from $\mathcal{C}$ to $\mathcal{C}'$. Assume that $\mathcal{C}$ has sufficiently many injective objects. For every object M of $\mathcal{C}$ there exist objects $R^i F(M)$ for $i \geq 0$ in $\mathcal{C}'$ satisfying the following conditions:

- (Conditions...)

3.

Definition (4.31). We say that an object M of $\mathcal{C}$ is *acyclic* for the functor F if we have $R^i F(M) = 0$ for all $i > 0$.

Proposition (4.32). Let $M^\bullet, i : A \rightarrow M^0$ be a resolution of A , where the M^i are acyclic for the functor F . Then $R^i F(A)$ is equal to the cohomology $H^i(F(M))$ of the complex $F(M^\bullet)$. (**Obs.** This cohomology of a complex is defined as a cokernel, def. 4.24)

4. (**Sect. 4.3**) From now on, we consider the category of sheaves of abelian groups over a topological space X , and the functor Γ of "global sections" which to $\mathcal{F}$ associates $\Gamma(X, \mathcal{F}) = F(X)$, with values in the category of abelian groups. The category of sheaves of abelian groups has sufficiently many injective objects. This implies the existence of the derived functors $R^i \Gamma$. They are written

$$R^i \Gamma(\mathcal{F}) =: H^i(X, \mathcal{F})$$

5.

Definition (4.33). A sheaf $\mathcal{F}$ is called *flasque* if for every pair of open sets $V \subset U$, the restriction map $\rho_{UV} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$ is surjective.

Proposition (4.34). Flasque sheaves are acyclic for the functor Γ .

6.

Definition (4.35). A *fine sheaf* $\mathcal{F}$ over X is a sheaf of $\mathcal{A}$ -modules, where $\mathcal{A}$ is a sheaf of rings over X satisfying the property:

For every open cover $U_i, i \in I$ of X there exists a partition of unity $f_i, i \in I, \sum f_i = 1$ (where the sum is locally finite), subordinate to this covering.

Proposition (4.36). If $\mathcal{F}$ is a fine sheaf, we have $H^i(X, \mathcal{F}) = 0 \forall i > 0$.

Corollary (4.37). Let X be a C^∞ manifold. Then

$$H^k(X, \mathbb{R}) = H_{dR}^k(X, \mathbb{R}).$$

(The left-hand side is the cohomology of the constant sheaf $\mathbb{R}$, and the right-hand side is the cohomology of the de Rham complex given by exterior differentiation of differential forms)

7.

Theorem (4.47). Let X be a locally contractible topological space. Then we have a canonical isomorphism

$$H_{\text{sing}}^q(X, \mathbb{Z}) \cong H^q(X, \mathbb{Z}).$$

The same result holds with $\mathbb{Z}$ replaced by any commutative ring G .

Proof.

- (a) Construct a complex of sheaves from singular cochains. This is done starting with the presheaf

$$U \mapsto C_{\text{sing}}^q(U, \mathbb{Z})$$

where $C_{\text{sing}}^q(U, \mathbb{Z})$ is just the dual of the q -chains, which in turn are just elements in the free group generated by simplices.

- (b) Such a complex is a resolution of $\mathbb{Z}$. (computation)
- (c) Such a resolution is acyclic for Γ . (prop 4.34; it is flasque)
- (d) By prop. 4.32 cohomology of acyclic resolution equals derived functors. **Constant sheaf $\mathbb{Z}$ is also acyclic resolution?**

$$H^q(X, \mathbb{Z}) = H^q(\Gamma(C_{\text{sing}}^\bullet))$$

- (e) $\Gamma(C_{\text{sing}}^\bullet)$ is quasi-isomorphic to $C_{\text{sing}}^\bullet$.

□

8.

Corollary (de Rham theorem).

$$H_{\mathrm{dR}}^q(X) \cong H_{\mathrm{sing}}^q(X, \mathbb{R})$$

Proof. Coro. 4.37 and thm 4.47 replacing $\mathbb{Z}$ with $\mathbb{R}$. □

Dolbeault's theorem

Here's an attempt to use the proof of de Rham's theorem as a model to prove Dolbeault's theorem!

Theorem (Dolbeault, coro. 2.6.21 Huybrechts). For a complex compact manifold M

$$H_{\bar{\partial}}^{p,q}(M) \cong H_X^q(M, \Omega^p)$$

Proof.

1. For every p we have a sheaf complex $\mathcal{A}^{p,\bullet}(X)$.
2. Such a complex is a resolution of the sheaf of sections $\Omega^{p,q}$. (**Obs.** See prop. 4.19 **voisin** to use instead the sheaf $\Omega^{0,q} \otimes E$, E holomorphic vector bundle.)
3. Such a resolution is acyclic. (maybe by Newlander-Nirenberg+Poincaré $\bar{\partial}$ -lemma?)
4. By **voisin** prop. 4.32 cohomology of this sheaf complex equals derived functors.
5. Dolbeault complex is also a resolution of the sheaf of sections $\Omega^{p,q}$.

□